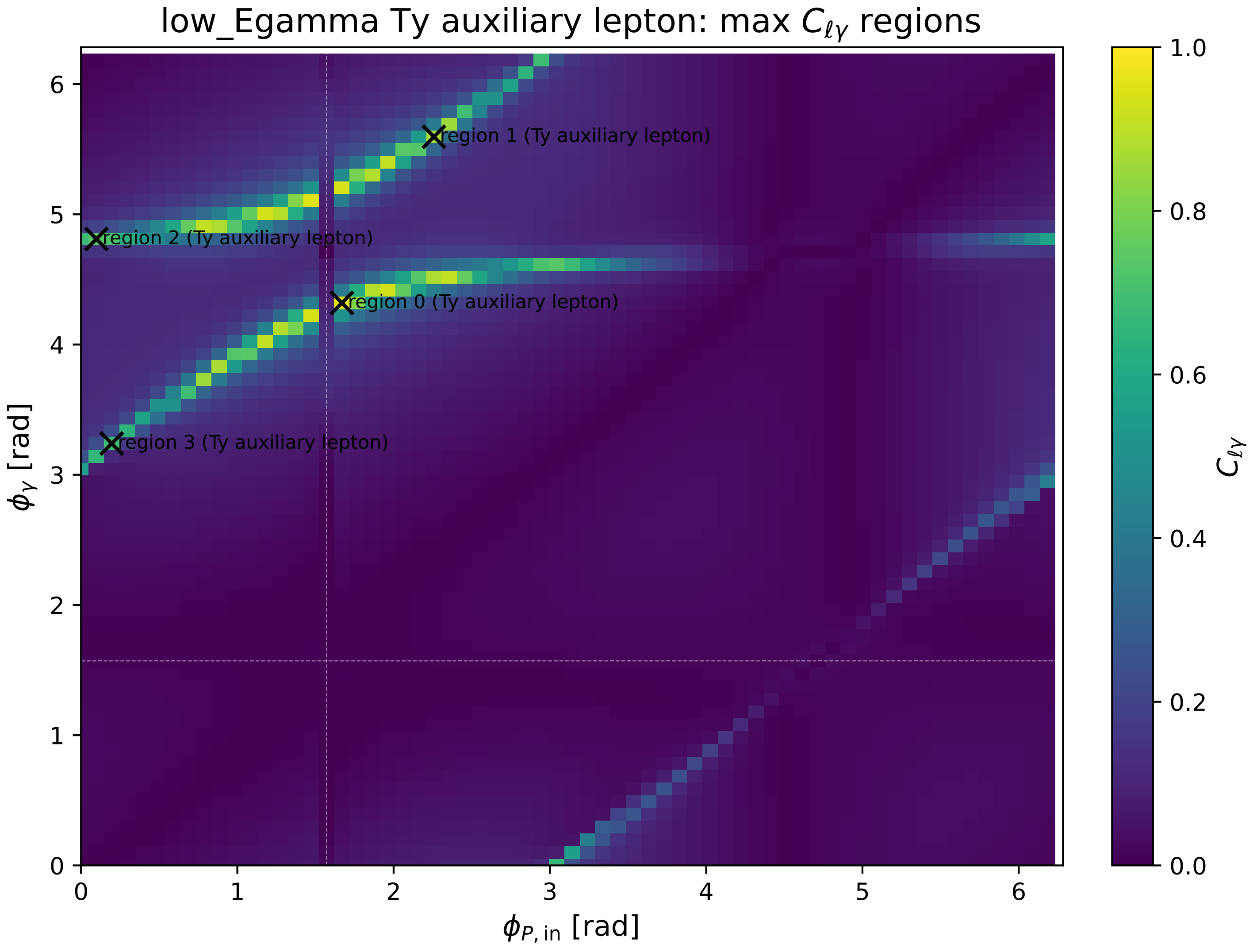
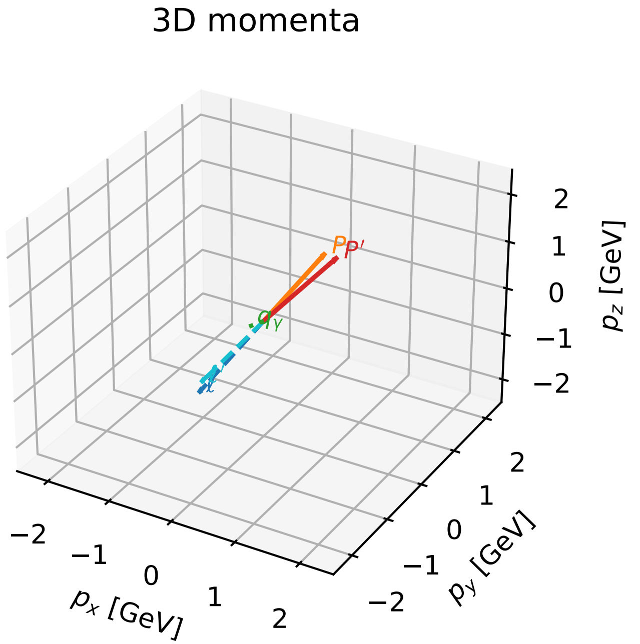


low_Egamma Ty auxiliary lepton: max $C_{\ell\gamma}$ regions
region 1 (Ty auxiliary lepton)
region 2 (Ty auxiliary lepton)
region 0 (Ty auxiliary lepton)
region 3 (Ty auxiliary lepton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\ell\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

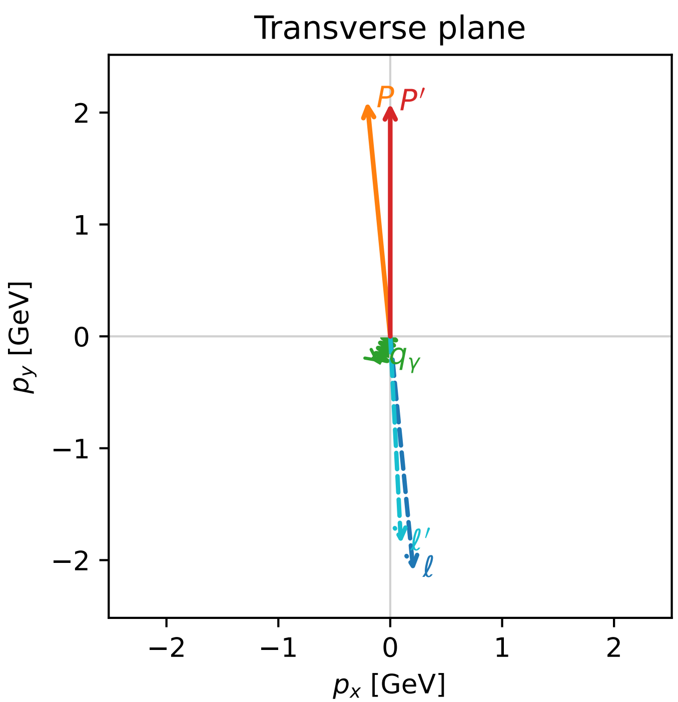
Ty auxiliary lepton characteristic max $C_{\ell\gamma}$ configuration

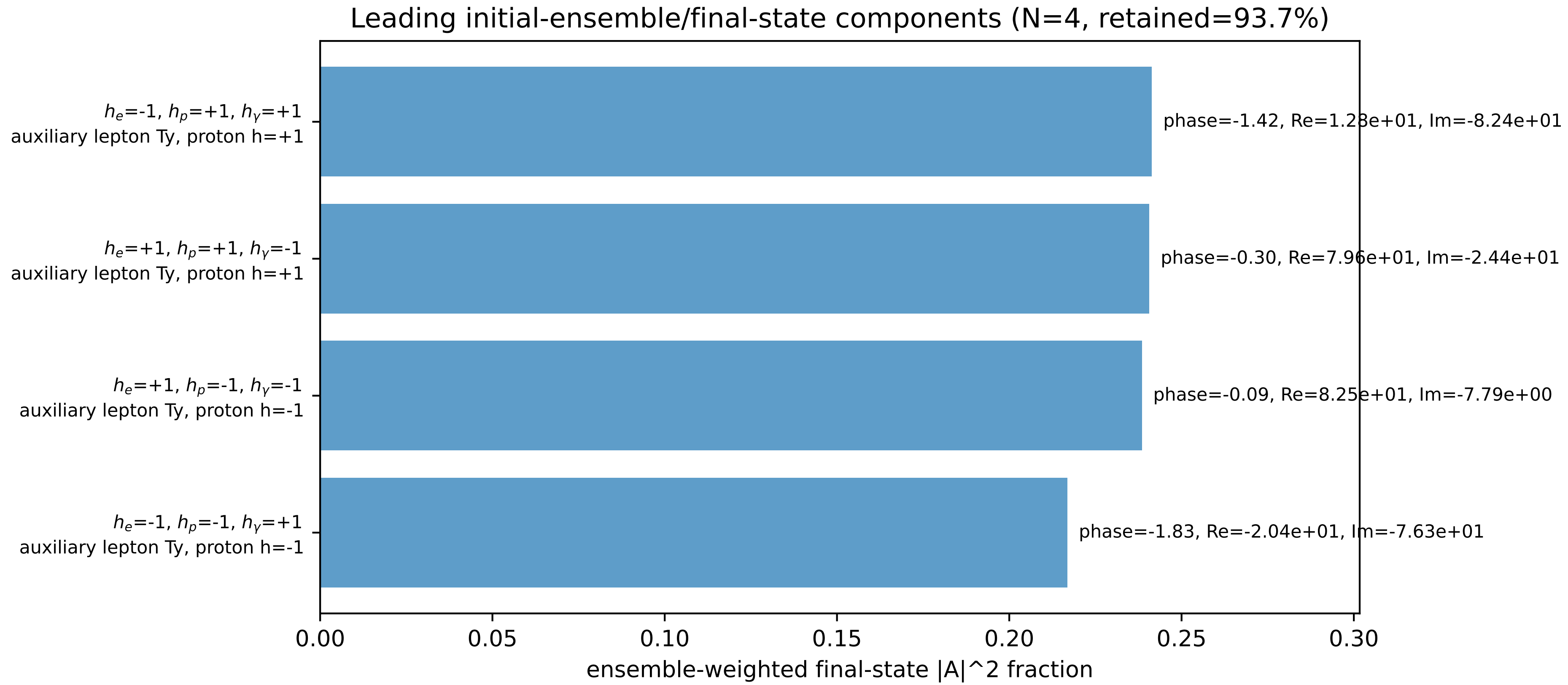


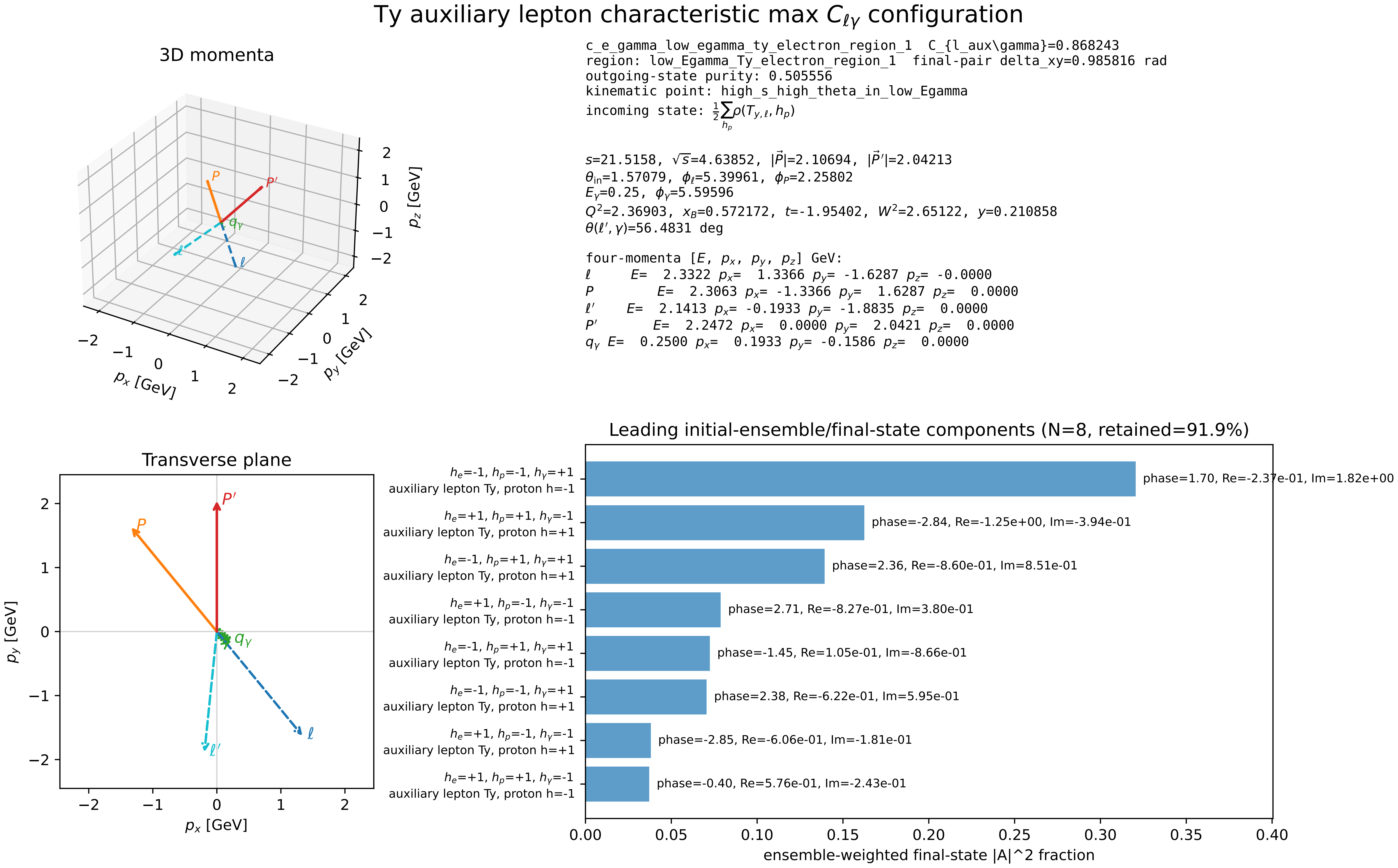
c_e_gamma_low_egamma_ty_electron_region_0 C_{l_aux\gamma}=0.949701
region: low_Egamma_Ty_electron_region_0 final-pair delta_xy=0.444355 rad
outgoing-state purity: 0.500746
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell},h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.0814
θ_{in}=1.57079, ϕ_ℓ=4.81056, ϕ_P=1.66897
E_γ=0.25, ϕ_γ=4.31969
Q^2=0.0215983, x_B=0.0101659, t=-0.0423427, W^2=2.98282, y=0.108198
$\theta(\ell',\gamma)$=25.4597 deg

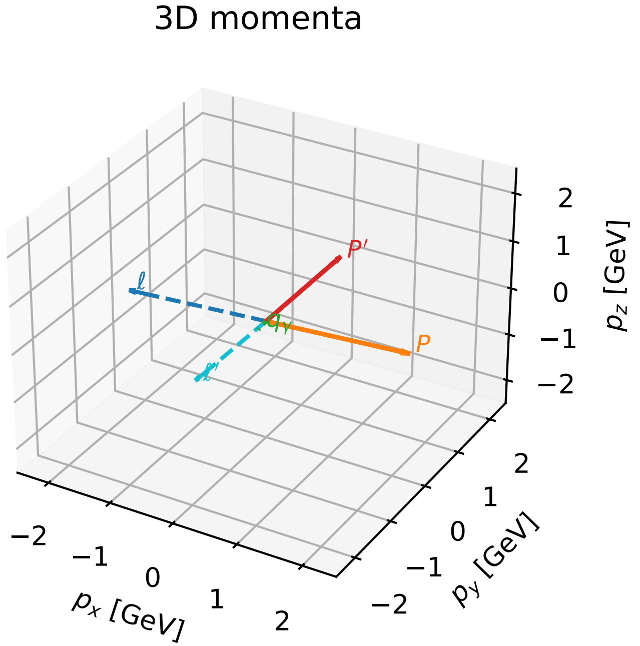
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 0.2065 p_y= -2.0968 p_z= -0.0000
P E= 2.3063 p_x= -0.2065 p_y= 2.0968 p_z= 0.0000
ℓ' E= 2.1055 p_x= 0.0957 p_y= -1.8504 p_z= 0.0000
P' E= 2.2830 p_x= 0.0000 p_y= 2.0814 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.0957 p_y= -0.2310 p_z= 0.0000






Ty auxiliary lepton characteristic max $C_{\ell\gamma}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
q_γ
ℓ'
ℓ
c_e_gamma_low_egamma_ty_electron_region_1 C_{l_aux\gamma}=0.868243
region: low_Egamma_Ty_electron_region_1 final-pair delta_xy=0.985816 rad
outgoing-state purity: 0.505556
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.04213
$\theta_{\rm in}$=1.57079, ϕ_ℓ=5.39961, ϕ_P=2.25802
E_γ=0.25, ϕ_γ=5.59596
Q^2=2.36903, x_B=0.572172, t=-1.95402, W^2=2.65122, y=0.210858
$\theta(\ell', \gamma)$=56.4831 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 1.3366 p_y= -1.6287 p_z= -0.0000
P E= 2.3063 p_x= -1.3366 p_y= 1.6287 p_z= 0.0000
ℓ' E= 2.1413 p_x= -0.1933 p_y= -1.8835 p_z= 0.0000
P' E= 2.2472 p_x= 0.0000 p_y= 2.0421 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1933 p_y= -0.1586 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=8, retained=91.9%)
h_e=-1, h_p=-1, h_γ=+1 auxiliary lepton Ty, proton h=-1
phase=1.70, Re=-2.37e-01, Im=1.82e+00
h_e=+1, h_p=+1, h_γ=-1 auxiliary lepton Ty, proton h=+1
phase=-2.84, Re=-1.25e+00, Im=-3.94e-01
h_e=-1, h_p=+1, h_γ=+1 auxiliary lepton Ty, proton h=+1
phase=2.36, Re=-8.60e-01, Im=8.51e-01
h_e=+1, h_p=-1, h_γ=-1 auxiliary lepton Ty, proton h=-1
phase=2.71, Re=-8.27e-01, Im=3.80e-01
h_e=-1, h_p=+1, h_γ=+1 auxiliary lepton Ty, proton h=-1
phase=-1.45, Re=1.05e-01, Im=-8.66e-01
h_e=-1, h_p=-1, h_γ=+1 auxiliary lepton Ty, proton h=+1
phase=2.38, Re=-6.22e-01, Im=5.95e-01
h_e=+1, h_p=-1, h_γ=-1 auxiliary lepton Ty, proton h=+1
phase=-2.85, Re=-6.06e-01, Im=-1.81e-01
h_e=+1, h_p=+1, h_γ=-1 auxiliary lepton Ty, proton h=-1
phase=-0.40, Re=5.76e-01, Im=-2.43e-01
ensemble-weighted final-state |A|^2 fraction

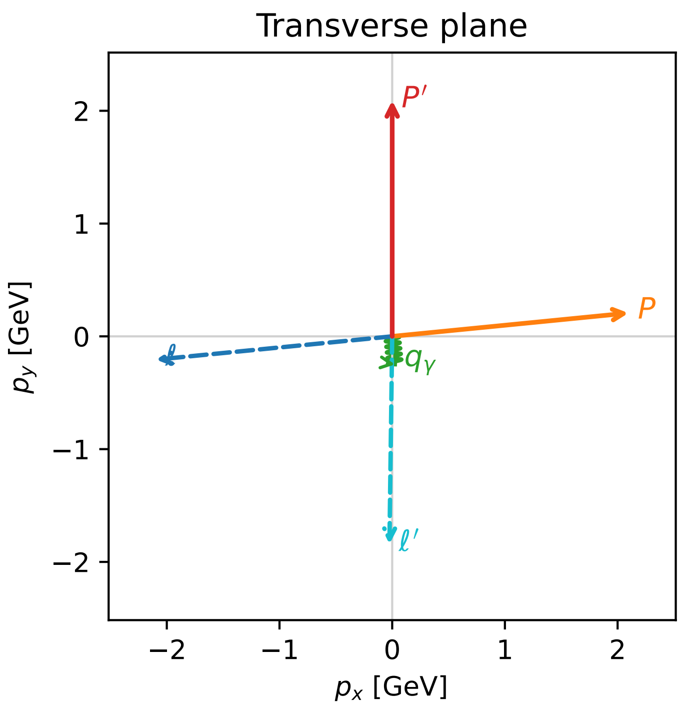
Ty auxiliary lepton characteristic max $C_{\ell\gamma}$ configuration

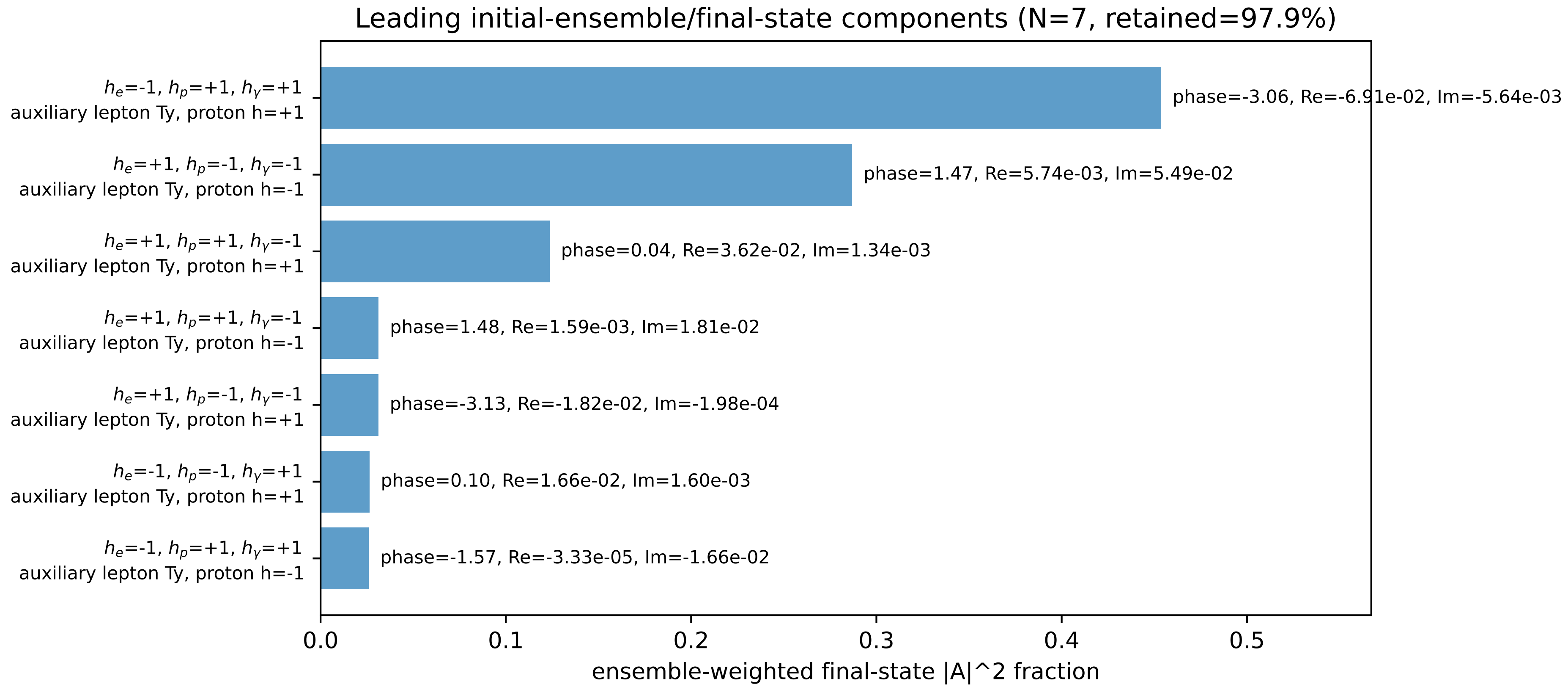


c_e_gamma_low_egamma_ty_electron_region_2 C_{l_aux\gamma}=0.734918
region: low_Egamma_Ty_electron_region_2 final-pair delta_xy=0.111474 rad
outgoing-state purity: 0.543446
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$

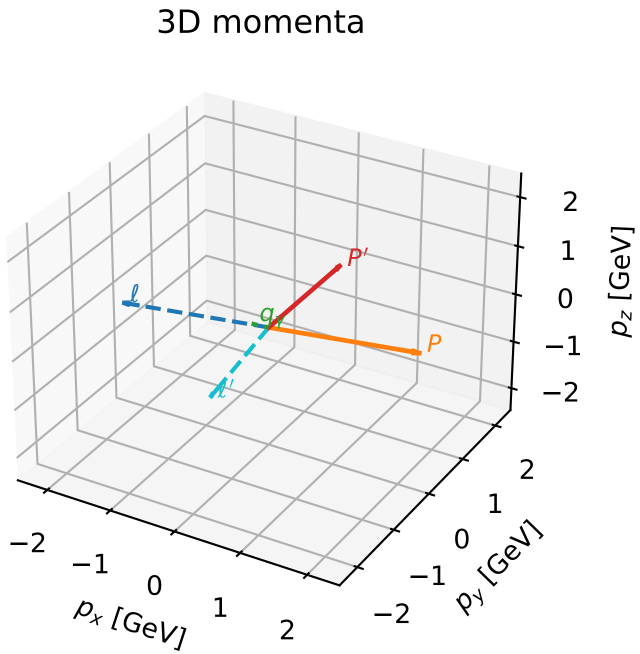
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.09128
θ_{in}=1.57079, ϕ_ℓ=3.23977, ϕ_P=0.0981748
E_γ=0.25, ϕ_γ=4.81056
Q^2=6.91524, x_B=0.759762, t=-7.94869, W^2=3.06645, y=0.463528
$\theta(\ell',\gamma)$=6.38697 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -2.0968 p_y= -0.2065 p_z= -0.0000
P E= 2.3063 p_x= 2.0968 p_y= 0.2065 p_z= 0.0000
ℓ' E= 2.0965 p_x= -0.0245 p_y= -1.8425 p_z= 0.0000
P' E= 2.2920 p_x= 0.0000 p_y= 2.0913 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.0245 p_y= -0.2488 p_z= 0.0000





Ty auxiliary lepton characteristic max $C_{\ell\gamma}$ configuration



c_e_gamma_low_egamma_ty_electron_region_3 C_{l_aux\gamma}=0.686869
region: low_Egamma_Ty_electron_region_3 final-pair delta_xy=1.59963 rad
outgoing-state purity: 0.529576
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.97282
θ_{in}=1.57079, ϕ_ℓ=3.33794, ϕ_P=0.19635
E_γ=0.25, ϕ_γ=3.23977
Q^2=7.7072, x_B=0.866355, t=-6.69456, W^2=2.06876, y=0.453051
$\theta(\ell', \gamma)$=91.6522 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -2.0665 p_y= -0.4110 p_z= -0.0000
P E= 2.3063 p_x= 2.0665 p_y= 0.4110 p_z= 0.0000
ℓ' E= 2.2041 p_x= 0.2488 p_y= -1.9483 p_z= 0.0000
P' E= 2.1845 p_x= 0.0000 p_y= 1.9728 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.2488 p_y= -0.0245 p_z= 0.0000